

SANTANDER, Spain

WMO: 080230

Lat: 43.4911N

Lon: 3.8006W

Elev: 194

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	39.2	40.9	28.7	22.8	44.8	31.5	26.0	45.6	38.8	48.7	34.8	50.1	10.2	150	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	77.8	67.3	75.3	67.4	73.6	67.0	71.1	74.6	69.8	73.2	68.6	72.0	11.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	110.4	73.5	68.4	105.2	72.0	67.3	101.1	71.0	35.0	74.8	33.9	73.4	32.9	72.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.0	26.6	23.0	35.8	86.2	2.6	4.1	33.9	89.2	32.4	91.6	30.9	93.9	29.0	96.8
			33.1	73.9	2.9	1.6	31.0	75.0	29.3	76.0	27.7	76.8	25.6	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	51.2	50.7	53.1	54.8	58.7	63.6	67.2	68.4	65.6	62.4	55.7	52.6	(6)
	DBStd	7.57	4.93	5.05	5.03	3.98	3.92	3.35	2.96	2.77	3.46	5.10	4.89	5.25	(7)
	HDD50	158	44	46	21	6	0	0	0	0	0	0	8	33	(8)
	HDD65	2593	428	401	370	307	199	64	12	4	32	111	280	385	(9)
	CDD50	3341	80	65	118	149	270	407	533	572	468	386	180	113	(10)
	CDD65	302	0	0	2	1	4	21	80	110	50	31	3	0	(11)
	CDH74	399	0	0	3	4	22	45	68	107	64	84	2	0	(12)
	CDH80	52	0	0	0	0	4	13	6	14	7	8	0	0	(13)
(14) Wind	WSAvg	10.1	11.7	11.3	11.2	10.4	9.6	8.7	8.2	8.3	8.6	9.7	11.8	11.4	(14)
(15) Precipitation	PrecAvg	44.4	4.2	3.4	3.3	3.8	2.8	2.4	1.7	2.7	3.7	4.0	6.4	5.2	(15)
	PrecMax	58.0	9.4	9.3	5.9	9.3	6.0	9.4	4.0	8.0	10.1	11.7	17.2	14.4	(16)
	PrecMin	29.5	0.2	0.3	0.0	0.8	0.8	0.5	0.5	0.5	0.6	0.5	1.4	0.4	(17)
	PrecStd	6.9	2.2	2.0	1.5	2.4	1.3	1.7	1.0	2.0	2.4	3.0	3.3	3.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	67.8	72.6	71.8	76.5	79.9	79.7	80.3	80.1	80.4	72.1	67.5	(19)
		MCWB	54.5	53.7	58.0	58.7	61.9	68.3	71.2	70.6	65.8	64.8	60.8	55.6	(20)
	2%	DB	63.2	63.6	67.2	66.4	69.3	72.6	75.5	76.4	75.2	76.1	68.3	64.7	(21)
		MCWB	53.3	52.9	56.0	56.7	60.6	66.1	69.8	69.8	67.2	63.0	58.4	54.7	(22)
	5%	DB	60.7	60.3	63.8	62.9	66.3	70.2	73.7	74.4	72.9	73.0	65.5	62.2	(23)
		MCWB	52.8	52.3	54.1	56.1	60.6	65.0	68.6	68.8	66.6	62.0	57.3	53.7	(24)
	10%	DB	58.2	57.5	60.2	60.5	64.2	68.4	72.1	73.1	71.0	69.9	62.8	60.0	(25)
		MCWB	51.7	51.5	53.5	55.8	59.5	64.1	67.5	68.0	66.0	61.9	56.6	52.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	57.0	59.3	61.1	65.1	69.6	72.9	73.7	71.1	67.6	63.2	58.4	(27)
		MCDB	61.7	63.1	68.0	66.5	71.0	75.4	77.0	78.5	74.8	72.7	68.0	62.8	(28)
	2%	WB	55.5	55.1	57.3	58.8	63.0	67.3	70.9	71.3	69.3	66.0	60.8	56.7	(29)
		MCDB	59.4	59.7	64.0	63.3	67.0	71.1	74.4	74.7	72.6	70.8	65.0	61.2	(30)
	5%	WB	54.2	53.8	55.8	57.5	61.6	65.8	69.5	70.0	68.1	64.7	59.2	55.3	(31)
		MCDB	58.1	57.7	61.3	61.2	65.1	68.8	72.8	73.4	71.1	69.5	63.1	59.8	(32)
	10%	WB	53.0	52.6	54.5	56.3	60.2	64.8	68.2	68.9	66.8	63.4	57.7	54.1	(33)
		MCDB	56.9	56.3	59.1	59.8	63.4	67.7	71.4	72.2	69.9	68.0	61.7	58.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	8.7	8.6	7.9	7.5	7.6	8.3	8.7	9.2	7.8	7.6	(35)
		MCDBR	10.2	11.9	14.3	13.1	12.1	10.4	10.0	10.4	11.3	13.7	11.4	10.3	(36)
	5% WB	MCWBR	6.3	7.3	7.6	7.5	6.8	6.5	6.4	6.4	7.0	6.8	6.5	6.3	(37)
		MCWBR	8.5	9.6	11.0	10.8	10.0	8.8	9.3	9.4	9.5	10.5	9.1	8.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.343	0.377	0.397	0.407	0.419	0.407	0.404	0.385	0.366	0.344	0.326	(40)
		taud	2.497	2.449	2.356	2.311	2.317	2.318	2.353	2.372	2.419	2.475	2.485	2.503	(41)
	Ebn at Noon		253	270	272	275	275	271	273	270	267	258	245	244	(42)
		Edh at Noon	23	28	35	39	40	40	38	36	32	27	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		436	685	1074	1437	1696	1820	1827	1632	1322	866	503	408	(44)
	RadStd		50	93	126	141	115	133	84	98	56	70	61	45	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page